A Two-Stage Hierarchical Multiclass Network Intrusion Detection System

with Tempered Class Weighting and a Reject Option for Imbalanced Network Flows

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1 INTRODUCTION

Machine-learning intrusion detectors often report accuracy above 99%, yet miss the rare attack families that matter most. Two causes recur: **severe class imbalance**, which accuracy hides, and **evaluation leakage** from duplicate records crossing the train–test split. This thesis builds a leakage-safe, imbalance-aware two-stage system that detects attacks and names their family on CIC-UNSW-NB15, judged by **macro-F1** rather than accuracy.

97.5% of full-release flows are benign

10,392 : 1 benign to Worms flows, after deduplication

26.4% of full-release records are duplicates

Research Objectives

- Build a **leakage-safe pipeline**: deduplicate first, fit every label-dependent step on training data only
- Compare **class-imbalance treatments** (class weighting, tempered weighting, SMOTE) on rare families
- Design a **two-stage gate + family ensemble** and test each component with paired bootstrap tests
- Evaluate at the **realistic 97.5% benign base rate** and under perturbed inputs

2 METHODOLOGY

The final system was trained on 2,085,442 flows and tested once on 521,361 held-out flows of the **full release (3.54M flows)**.

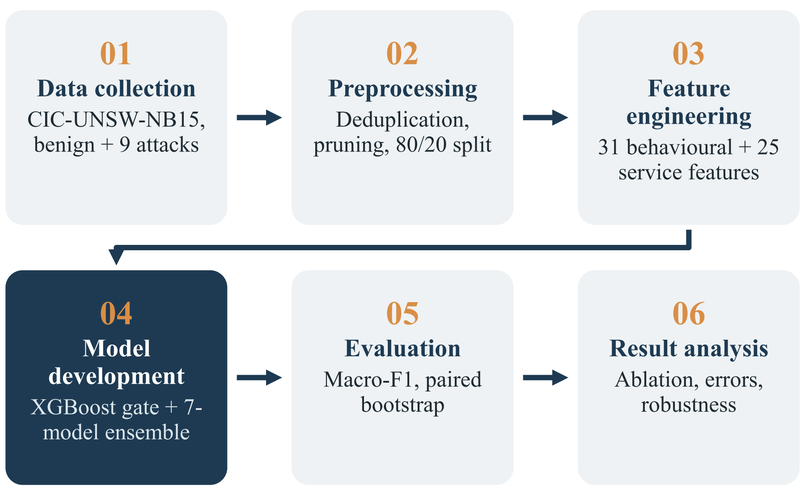


Figure 1: Research flow, from data collection to result analysis

3 DATASET & PREPROCESSING

CIC-UNSW-NB15 (2024) re-extracts UNSW-NB15 traffic with CICFlowMeter: **76 flow statistics** per record. Steps that use labels run after the split, on training data only.

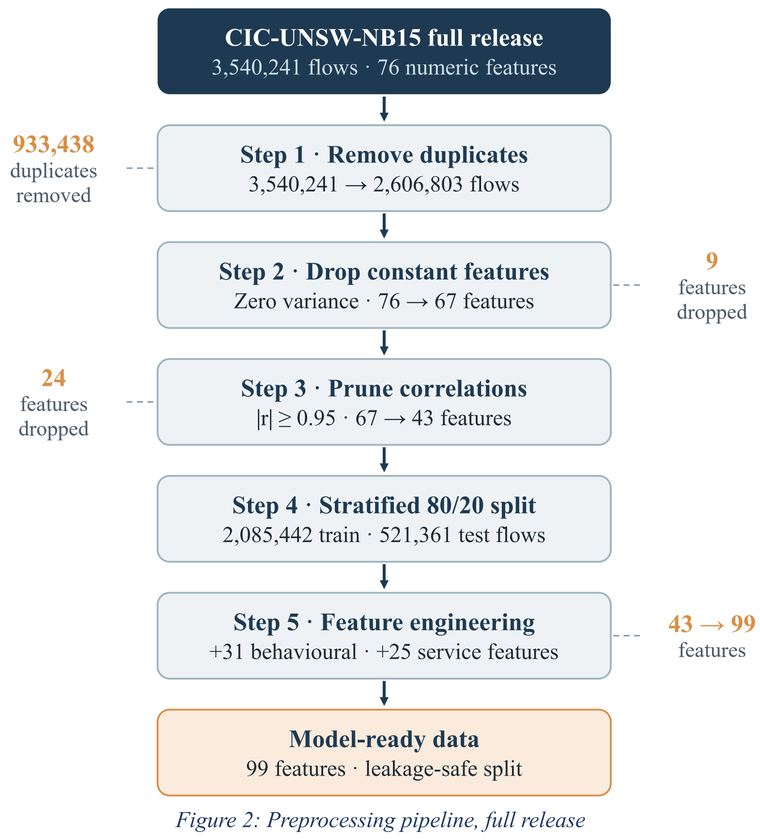


Figure 2: Preprocessing pipeline, full release

Attack flows per family, full release (after deduplication)

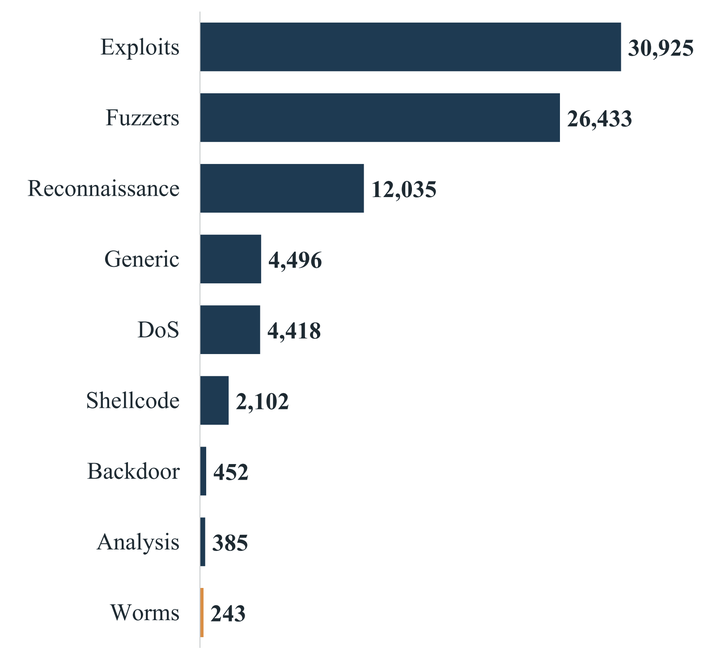


Figure 3: Class imbalance. Worms has 243 flows against 2,525,314 benign

4 SYSTEM ARCHITECTURE

A binary **gate** screens every flow; only suspected attacks reach a **seven-member ensemble** that names the family. Members differ in how strongly they rebalance classes.

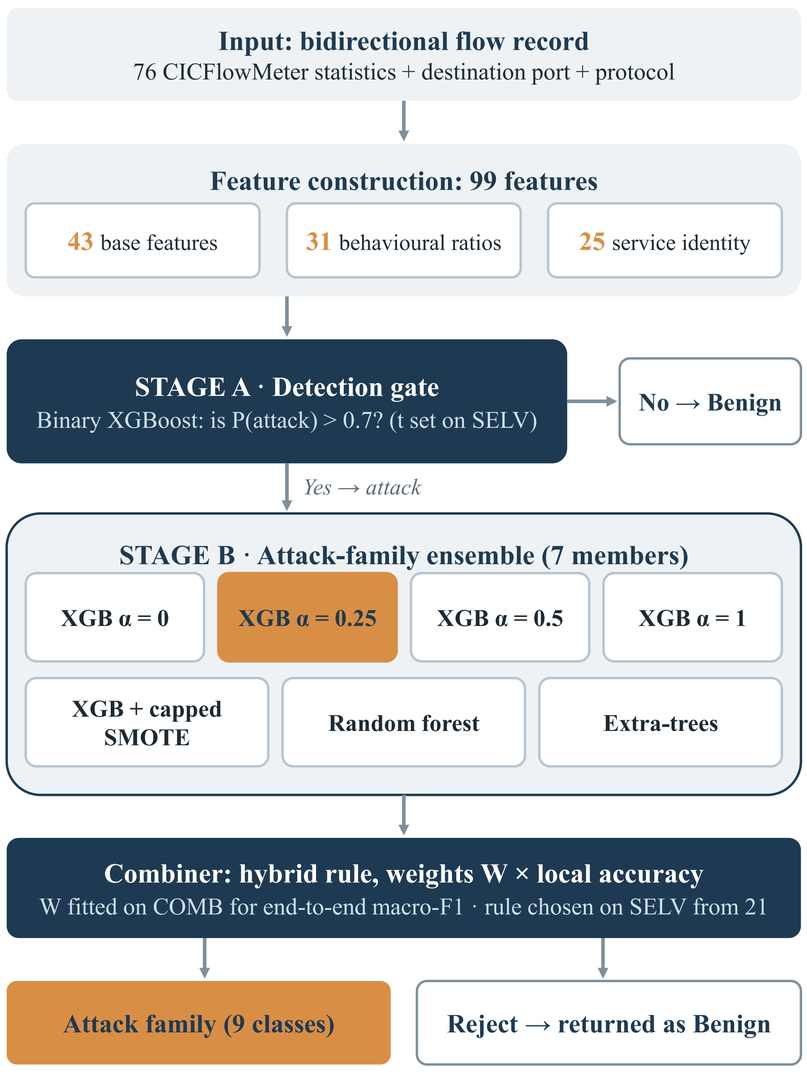


Figure 4: Architecture; COMB and SELV are the two halves of the training data

Leakage-safe Protocol

- Duplicates removed **before** the train–test split
- Combiner weights fitted on **cross-fitted out-of-fold** predictions
- Test set scored **once**; every gain checked with a **paired bootstrap** (2,000 resamples)

5 RESULTS

0.689 macro-F1, full release [0.672, 0.704]

98.66% accuracy, full-release test set

2.6× fewer benign false alarms

Macro-F1 by configuration, full-release test set (521,361 flows)

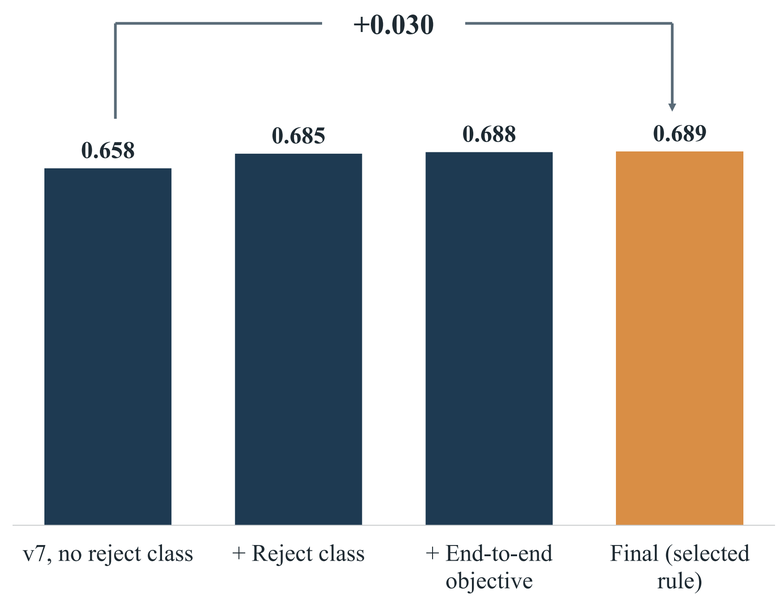


Figure 5: The reject class gave the largest gain (+0.027, $p < 0.001$)

Per-class F1, final system, full-release test set (521,361 flows)

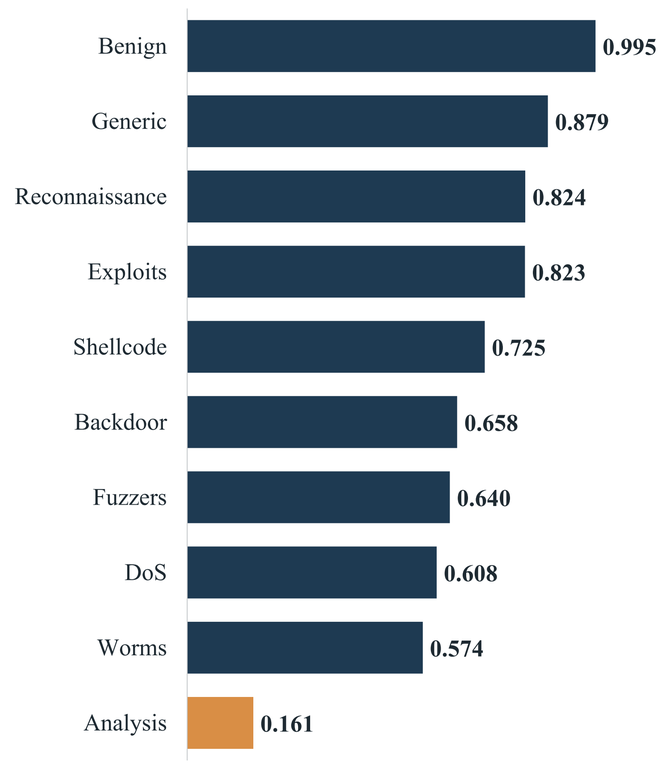


Figure 6: Analysis is the weakest family; 47 of its 77 test flows were labelled benign

6 ANALYSIS & GENERALIZATION

Gains came from **better information and moderate rebalancing**, not from a bigger model.

Macro-F1 gain per component, paired bootstrap

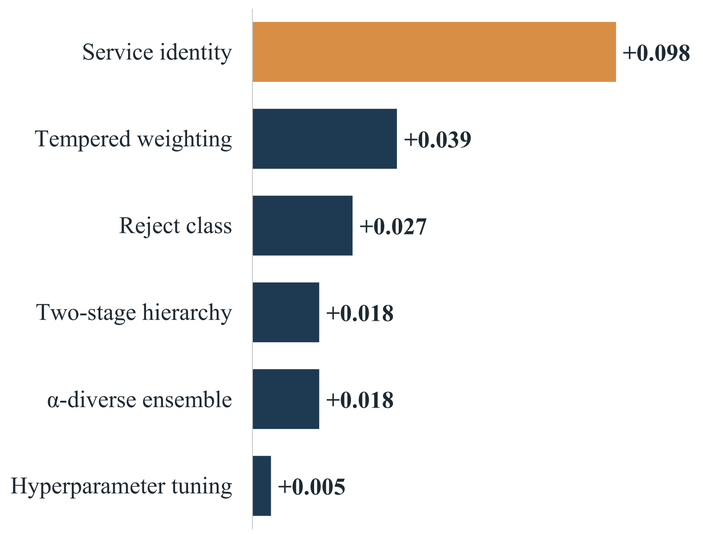


Figure 7: Hyperparameter tuning (+0.005) was not significant ($p \approx 0.23$)

- Errors come from **class overlap**, not label noise: only 0.35% of flows have conflicting labels
- k-NN overlap tracks model confusions (Spearman **0.827**); Exploits and Fuzzers were confused 1,287 times

Results on Other Datasets

Dataset	Macro-F1	Accuracy	Key observation
CIC-UNSW-NB15 full release (this work)	0.689	0.987	Main result, 97.5% benign
UNSW-NB15, original release	0.639	0.830	Service fields dominate
CIC-IDS2017, 13 classes	0.984	0.999	Too easy: 1-NN gets 99.7%
TON_IoT, network subset	0.960	0.981	Source IP alone flags 99.98% of attacks

Table 1: The same pipeline run on three other datasets

- On the original UNSW-NB15 release, **service fields** again carried most of the predictive power, confirming the service-identity result
- High scores on CIC-IDS2017 and TON_IoT come from **easy testbeds**, not a better model, so they cannot serve as fair second tests

7 CONCLUSION

- Final system: **macro-F1 0.689** and **98.66% accuracy** at the realistic 97.5% benign base rate
- **Moderate rebalancing** (α = 0.25, +0.039) beat full SMOTE and "balanced" weights
- **Information beat model capacity**: service identity added +0.098; tuning gave no established gain
- **Limitations:** single testbed, flow features only, weak Analysis and Worms detection
- **Future work:** payload features, cross-network tests, SHAP, adversarial training

8 REFERENCES

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